香港教育大學
The Education University of Hong Kong

MSc(AIET) 2025-26

MTH6184 Data Mining in STEM Education

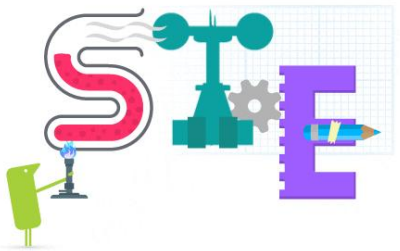
Chapter 7. STEM Lesson Design

Instructor: Philip Yu

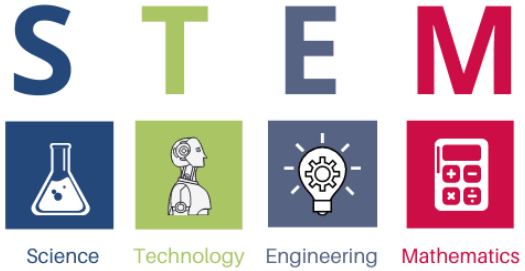
Chapter Outcomes

After completing this chapter, you are able to:

- Understand the Role of STEM Education
- Understand different approaches to create STEM lessons
- Understand the basic steps of designing a STEM lesson plan

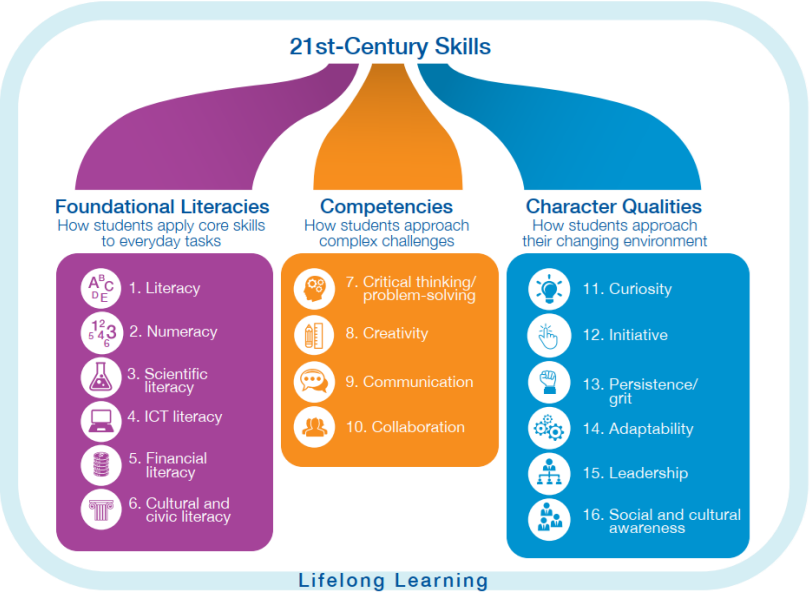


What is STEM?



https://www3.weforum.org/docs/WEF_New_Vision_for_Education.pdf

Exhibit 1: Students require 16 skills for the 21st century

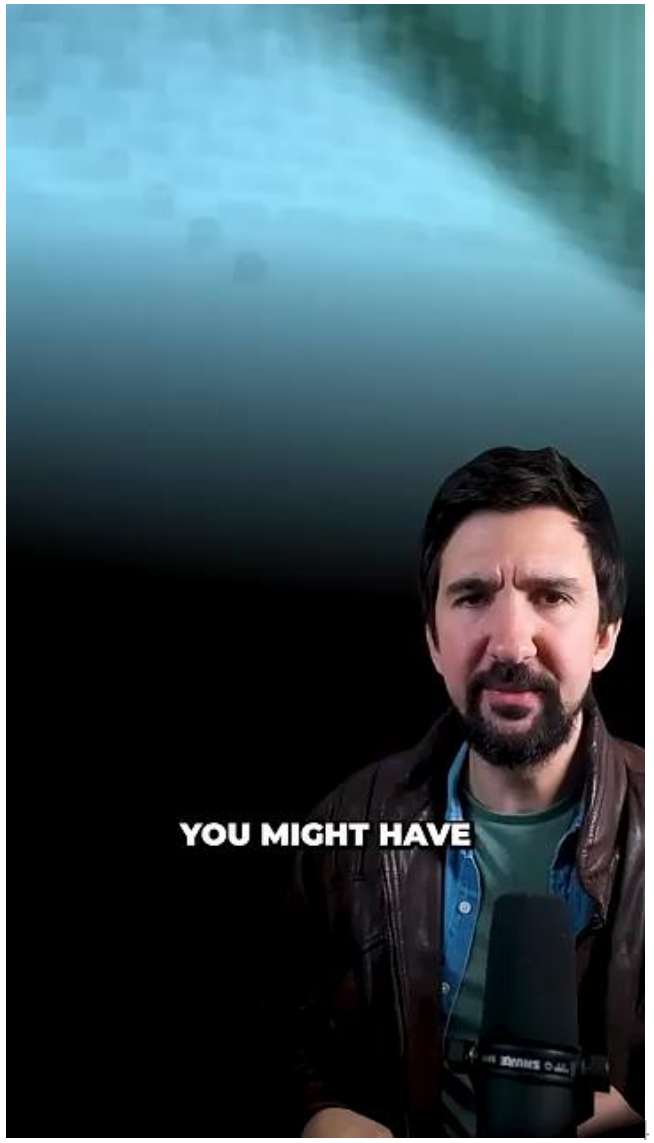


- No one doubts that science and technology play a major role in developing our 21st century skills.
- In 2001, an educational movement has arisen that aims to introduce young students to the various aspects of STEM (Science, Technology, Engineering, Mathematics).
- STEM is an acronym for science, technology, engineering, and mathematics. It is an integration of science, technology, engineering, and mathematics into a new cross-disciplinary subject in schools.
- Devised in 2001 by Dr. Judith Ramaley, the term has won widespread acceptance as a convenient way of referring to these academic disciplines.
- In November 2009, President Barack Obama launched the Educate to Innovate initiative, securing over US$1 billion in private investment for STEM education.
- Hong Kong's 2015/16 policy addresses pledged to renew STEM curricula and train teachers, providing one-off grants to all primary and secondary schools. In 2019, HK$500 million funded an IT lab program for secondary schools, extended to primary schools this year with over HK$200 million. https://www.thestandard.com.hk/opinion/article/1386/

Dr. Judith Ramaley (born 1941) is an American biologist and academic administrator coined the STEM acronym in 2001, when she served as director of the National Science Foundation's Education and Human Resources Division.

The Need for STEM Education

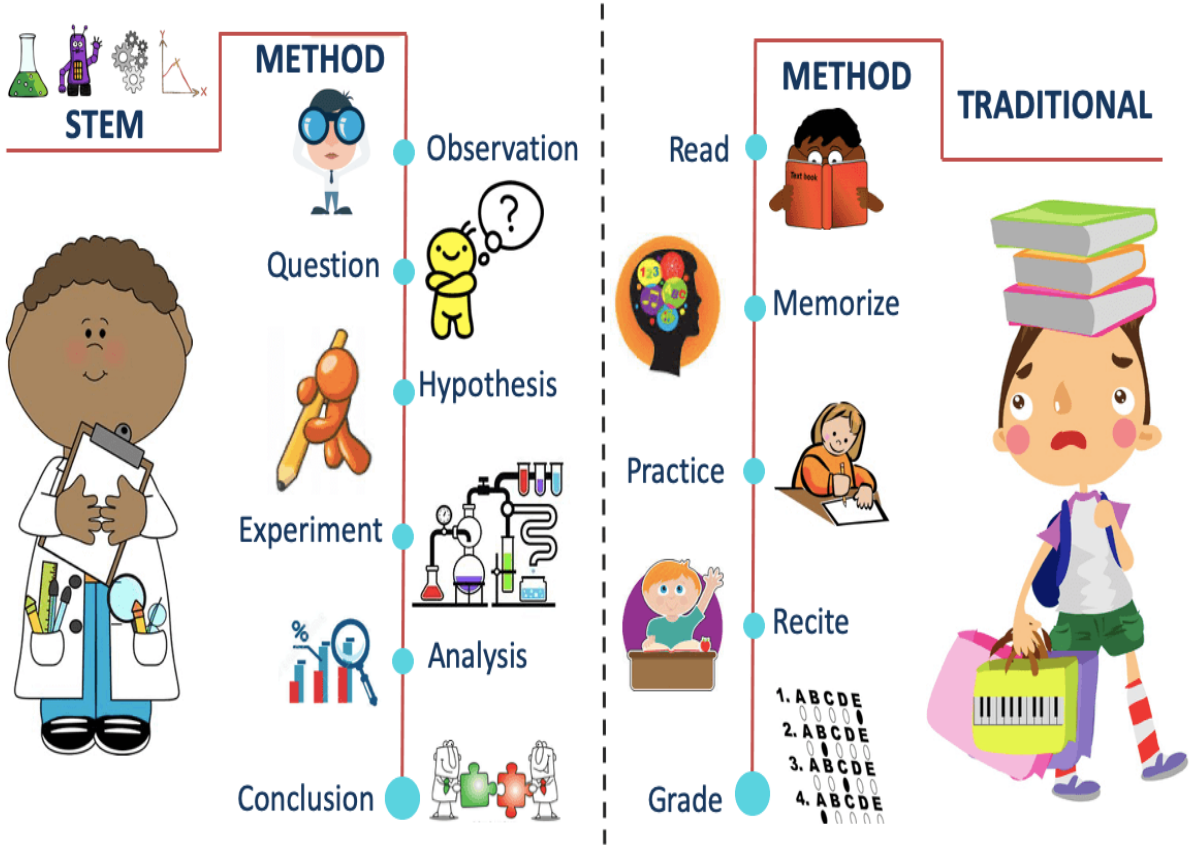
Scientists Create Shapeshifting Liquid Robots
Paper: Science Advances March 2025



- The demand for reliable workers with STEM-related skills will only increase with time.
- According to the National Science Foundation, approximately 80% of jobs that will be created over the next decade will demand the use of math and/or science skills.
- SKILLS GAP: short supply of highly-skilled workers in technological jobs and because of that hundreds of thousands of positions are going to be unfilled in the near future and is going to be a severe threat to businesses all over the world.
- Some experts believe the skills gap will worsen over the next 10 years, and suggest that educational institutions should train more students in STEM subjects: science, technology, engineering and mathematics.

STEM Education vs. Traditional Education

- Traditionally it has been the practice to teach the four STEM components more or less separately from one another.
- Instead of teaching four disciplines separately, STEM integrates them into a cohesive learning paradigm based on real-world applications.
- STEM education emphasizes student exploration and problem solving as opposed to rote learning.

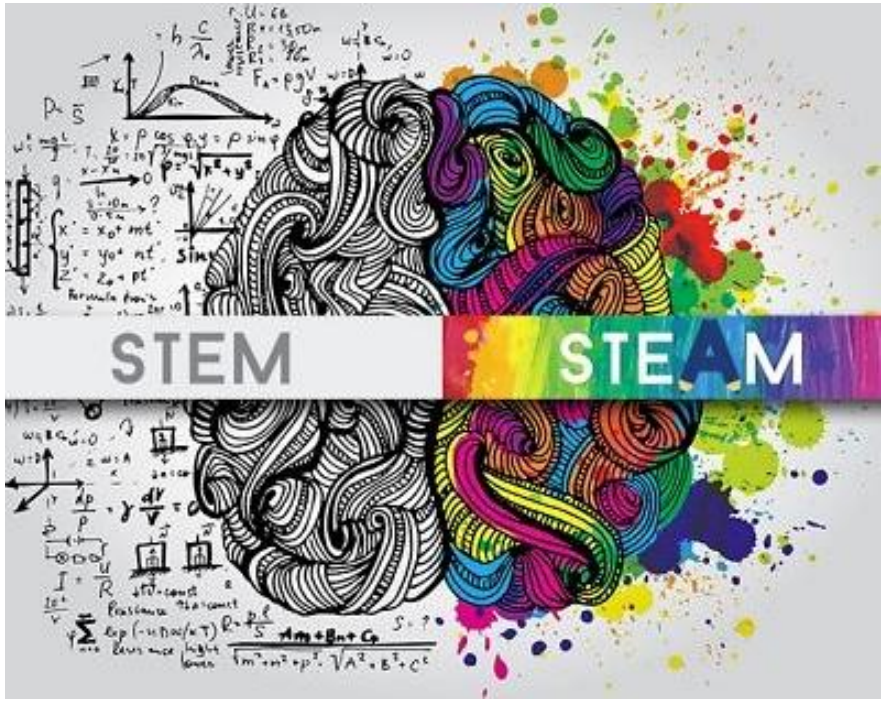


STEAM – Inclusion of ARTS

- The inclusion of Arts in the already existing STEM approach is a step ahead in the education. Using STEAM ('A' refers to Arts) methodology, a teacher connects critical areas of learning with art practices and design principles for holistic development. Such methodology exposes the students to new domains and removes limitations of innovation and creativity.

- Arts education creates focused, well-rounded and creative students that can take on any challenge.

- Put simply, the students start looking at the science and engineering problems from an artistic point-of-view and deliver the solution accordingly.



https://online.uwa.edu/infographics/steam-education/

Hong Kong STEAM Education Timeline

- 2015 - Published “Ongoing Renewal of the School Curriculum: Focusing, Deepening and Sustaining“, making the promotion of STEM education a key development focus of curriculum ongoing renewal.
- 2016 - Released the report on “Promotion of STEM Education – Unleashing Potential in Innovation“, outlining the development direction and recommended strategies for promoting STEM education.
- At the end of 2020, the Curriculum Development Council established the "Standing Committee on STEM Education," responsible for overseeing the overall development direction and support strategies for STEAM education (Note: later renamed the "Standing Committee on STEAM Education").
- The Chief Executive's 2022 Policy Address emphasized promoting STEAM education in primary and secondary schools for all (普及化), diversity (多元化), and fun (趣味化).
- The Chief Executive's 2023 Policy Address mentioned further promoting STEAM education in primary and secondary schools.
- The Chief Executive's 2024 Policy Address mentioned that the Education Bureau would establish a "Steering Committee on Strategic Development of Digital Education" to promote digital education, update the junior secondary science curriculum, and support teachers in using artificial intelligence for teaching.

Class Activity 1:

STEM Elements in A Lesson

STEAM: Designing a Catapult for Ancient Egypt

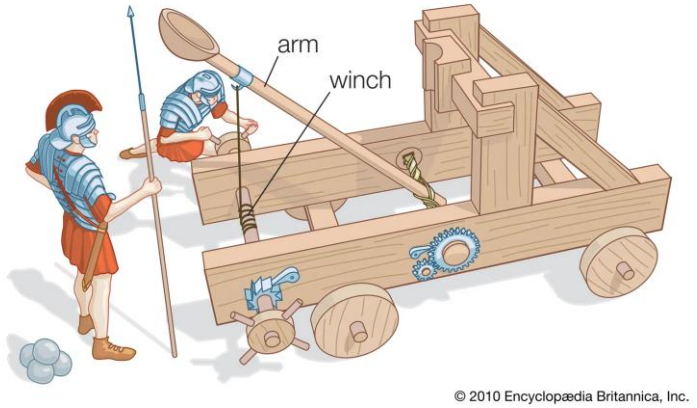
Level: Secondary 1

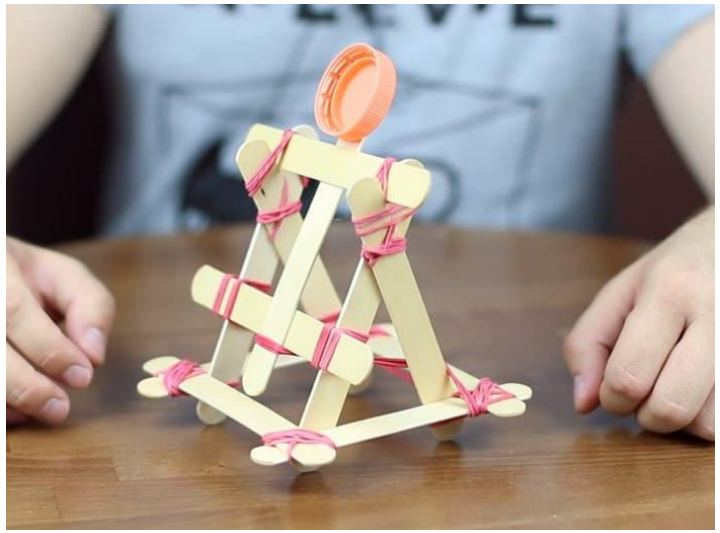
Topic: Energy

Learning Skills:

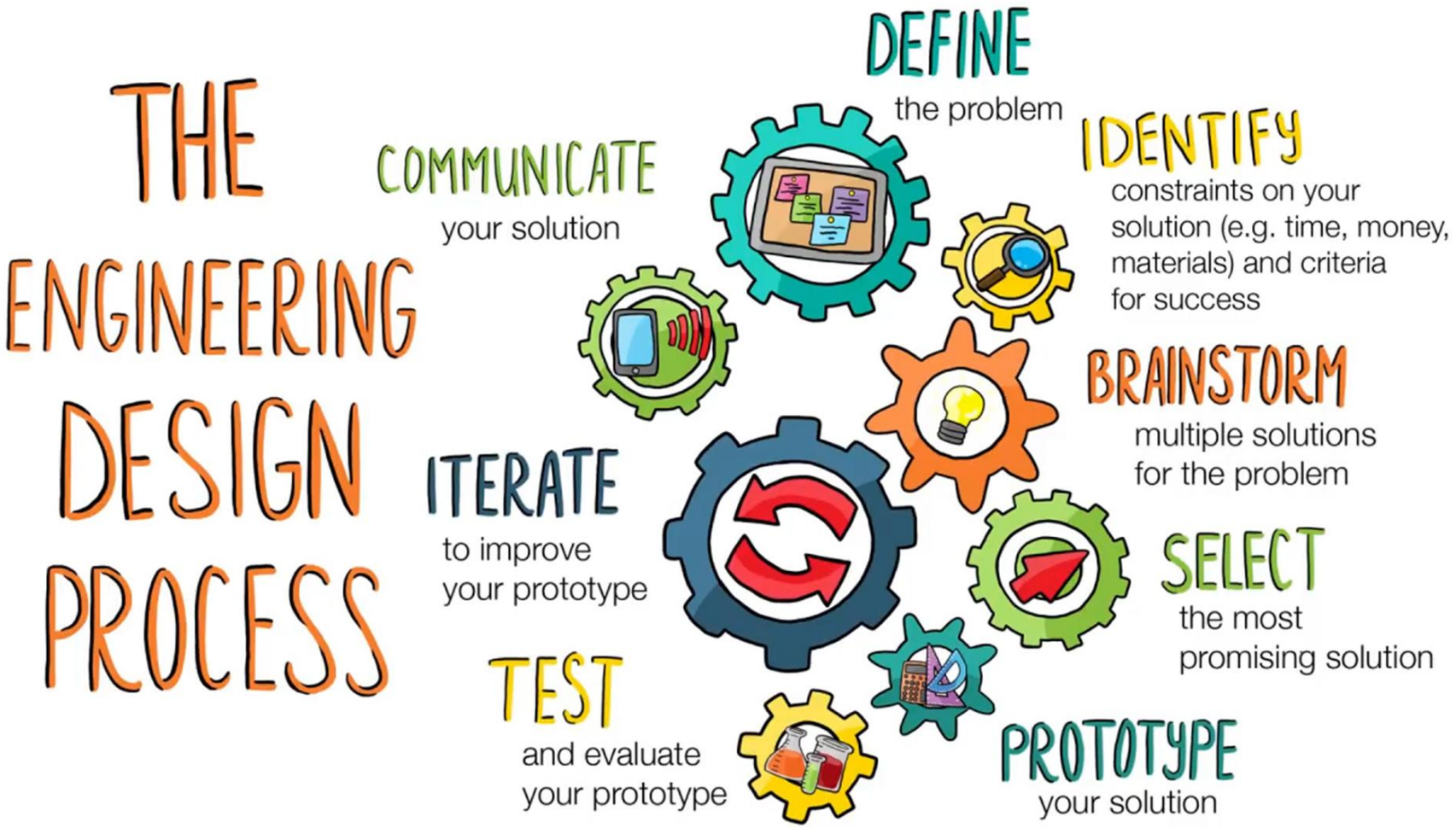
- Science:
-
- Technology:
- Engineering:
- Arts:
- Mathematics:

- **Background:** Catapults were first put to military use in ancient times. Ancient Egyptians used the device to hurl stones and flaming coals at enemy positions and installations.





STEAM: Designing a Catapult for Ancient Egypt



Reference: The Engineering Design Process Taco Party: https://education.nationalgeographic.org/resource/engineering-design-process-taco-party/

Designing a Model Rocket

Level: Secondary 1 to 3

Topic: Science: Force and Motion

Learning Skills:

- Science:
- Technology:

- Engineering:
- Mathematics:

Background: Inspired by NASA's rocket launches sending astronauts to the International Space Station, you will learn how to create your own model rocket using simple materials. Get ready to explore the science of propulsion, experiment with (water/air) pressure, and see how high your rocket can soar!

STEM: Identifying Vehicle CO2 Emission Factors

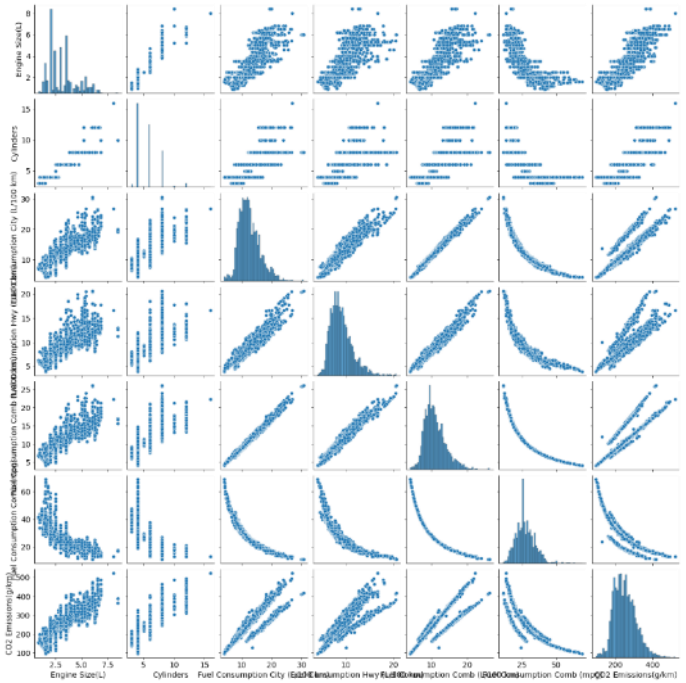
Level: University

Topic: Air Pollution and Energy

Learning Skills:

- Science:
- Technology:
- Engineering:
- Mathematics:

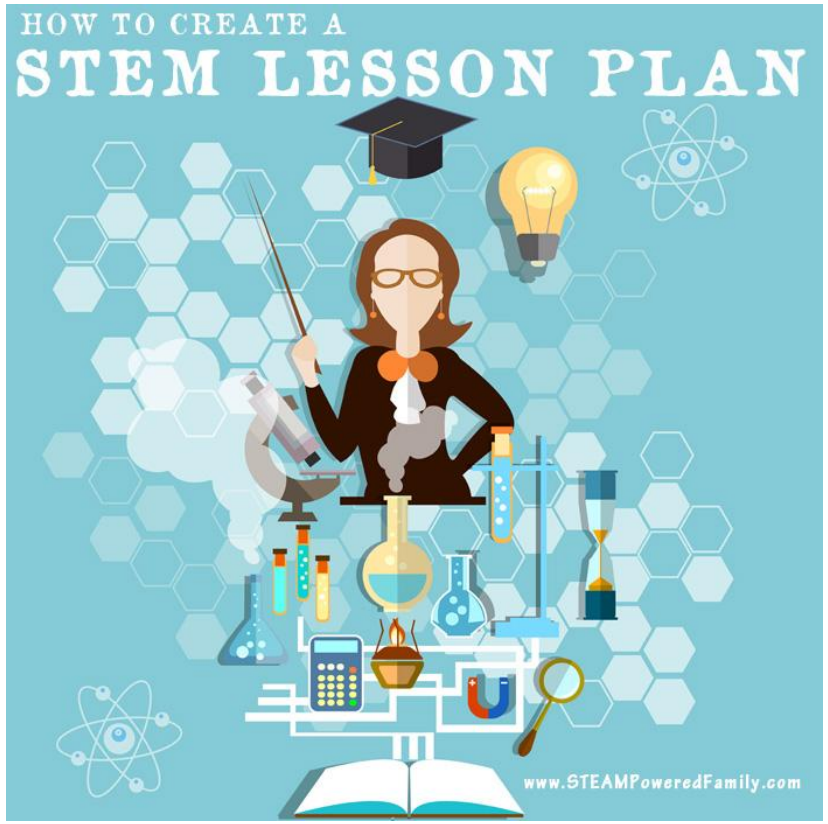
Background: Carbon dioxide is a kind of greenhouse gas that contributes to climate change. Greenhouse gas emissions have caused the global warming in the past ten years. Transport is one of major sources of carbon emissions. It would be of great interest identify the most influential features of a vehicle that affect the CO2 emission.



How to Design a STEM Lesson Plan?

A STEM Lesson Plan

- STEM lesson plans usually consist of longer term projects where students are doing the majority of the work and teachers are acting as a facilitator.
- Through hands-on, problem-solving interdisciplinary learning activities both inside and outside the classroom (including scientific inquiry, project-based learning, and design and production), students can comprehensively apply STEM knowledge and skills, stimulating creative thinking and cultivating STEM literacy to solve everyday problems.
- The implementation of STEM education emphasizes overall school planning and interdisciplinary collaboration.
- It also aims to discover and cultivate students with STEM potential.
- A STEM teacher is responsible to
  - create lesson plans that teach the necessary skills and concepts and also engage and excite your students.
  - guide students take the lead working through the project.



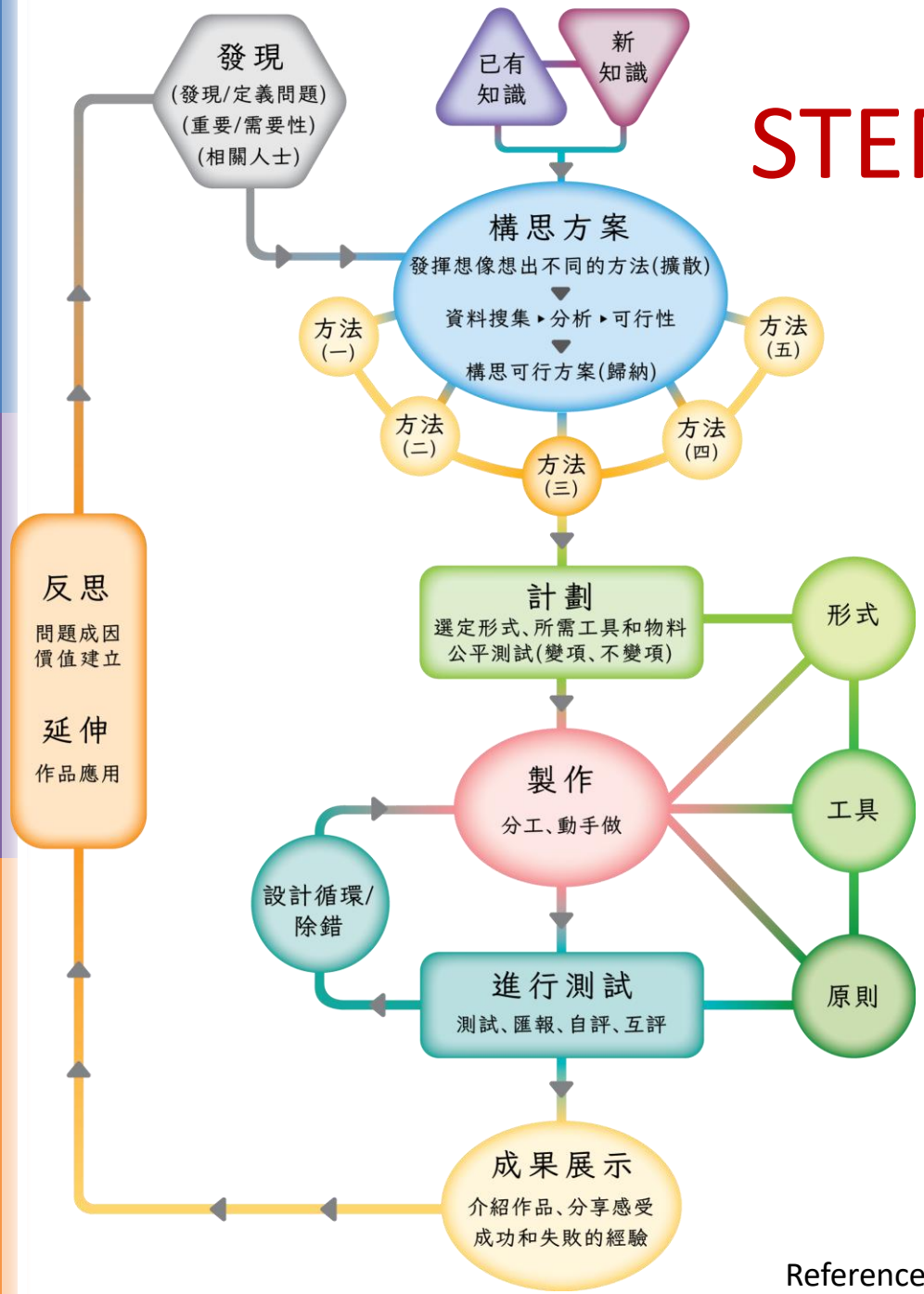
Basic Steps of Designing a STEM Lesson Plan

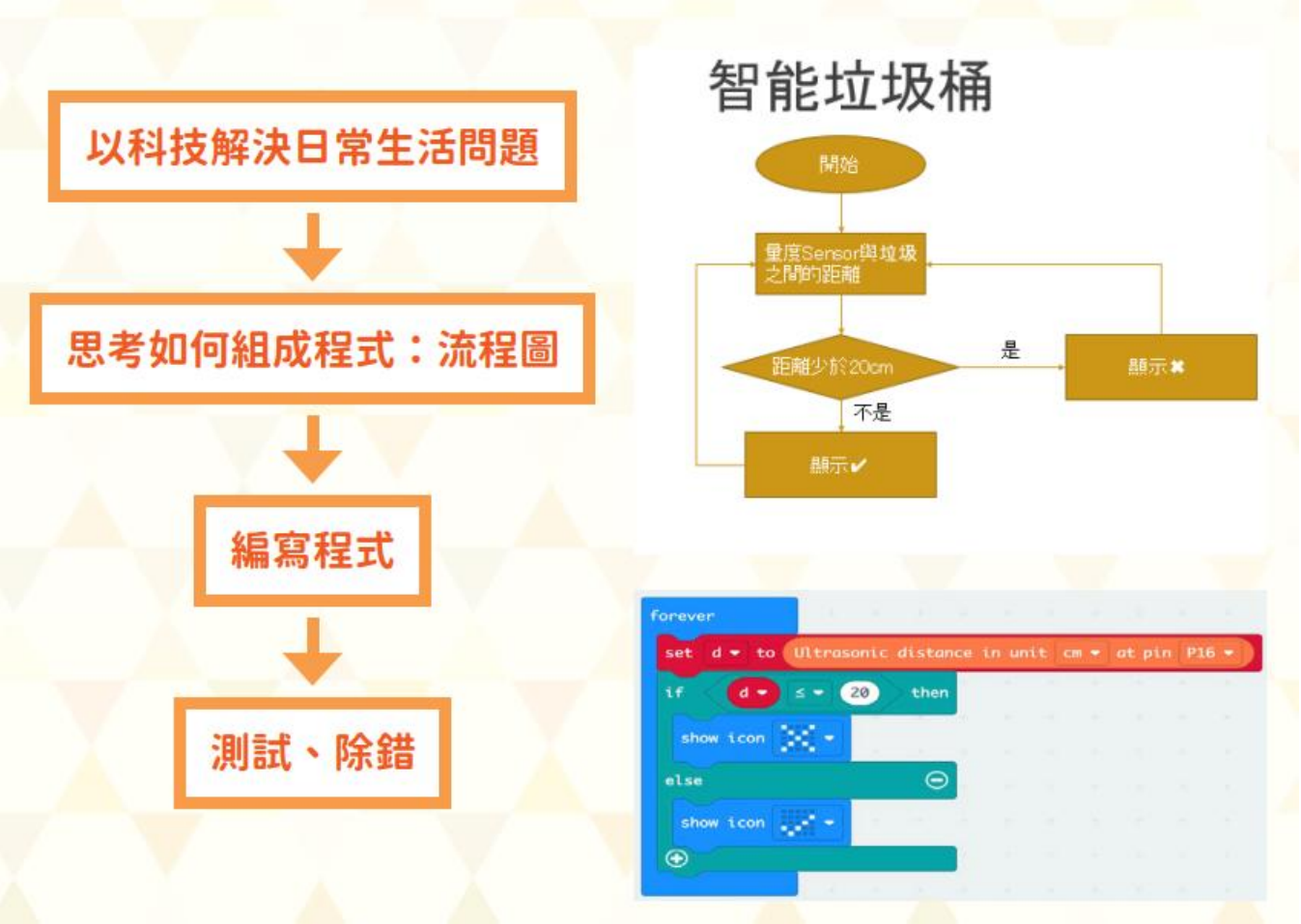
1. Choose a broad STEM topic and specify the objective.
2. Specify the learning outcomes.
3. Design a series of activities using an experiment, software package, simulated platform, online platform, video, reading material, etc.
4. While assigning an activity to the students, ask the students to collect some of their key observations on the activity sheet. What did they notice? Is there anything they found surprising?
5. Discuss each activity conducted in small groups or as a class.
6. Summarize the different learnings and other outcomes of each activity.
7. Based on each activity sheet, assessment criteria could be included at the end of activity sheet by using rubrics or quizzes to check does the final outcomes of the activity align with the pre-specified learning outcomes.

Online Platforms for implementing STEM Lessons:

PhET INTERACTIVE SIMULATIONS | CoSpaces EDU | Teachable Machine | Scratch A.I. | AI WITH APP INVENTOR

STEM Lesson Activities Workflow





Reference: 學校整體課程規劃與STEM教育(三) (書本) https://rdsys.edcity.hk/admin_file/64805

STEM Lesson: Smart Trash Can 智能垃圾桶

教學設計：

教節	學習重點	教學流程【教學活動、小結、總結及評估等】	資源 / 教具
1	瞭解學校校工叔叔清理垃圾桶時的辛勞	（一）熱身活動 (高階思維工作紙 – 全面因素) 播放有關校工叔叔清理垃圾的片段，了解校工叔叔在清理垃圾時所遇到的問題。 (利用 Ipad Mentimeter 收集學生的意見 / 以全面因素圖像填寫意見)	每組獲派： 高階思維工作紙 – 全面因素
		（二）導入 播放坊間智能垃圾桶的影片，讓學生對智能垃圾桶有基本的了解。 老師引導學生討論： ➔ 以上的裝置，能否解決校工叔叔清理垃圾時所遇到的問題呢？(不能) ➔ 垃圾桶要怎樣設計才可解決校工叔叔的問題 呢？(學生思考)	
		（三）教學流程 解決問題方案： **提問**：智能垃圾桶應具備什麼功能，才能方便校工叔叔清理呢？ (利用 Ipad Mentimeter 收集學生的意見) (估計學生回答 : 亮燈、顯示垃圾桶已滿、垃圾桶快滿時發送信息給校工叔叔)	
		（四）智能垃圾桶展示 老師向學生展示一個已安裝了 micro：bit 的智能垃圾桶，下一節課，學生需就校工叔叔的需要而進行改良。	
		（五）總結 科技的進行可讓生活更方便。	

Remarks

- Find or create supplemental materials your students will need.
  - Effective STEM lesson plans may require supplemental materials like videos, research articles and online demos to guide students.

- Plan how you will encourage collaboration and teamwork.
  - Effective STEM learning thrives on collaboration and teamwork, so incorporate activities like group roles, reflection, and peer feedback to foster a cooperative environment.
  - This intentional planning minimizes conflicts and equips students with essential teamwork skills for future STEM careers.

Integrating Data Mining into STEM Education

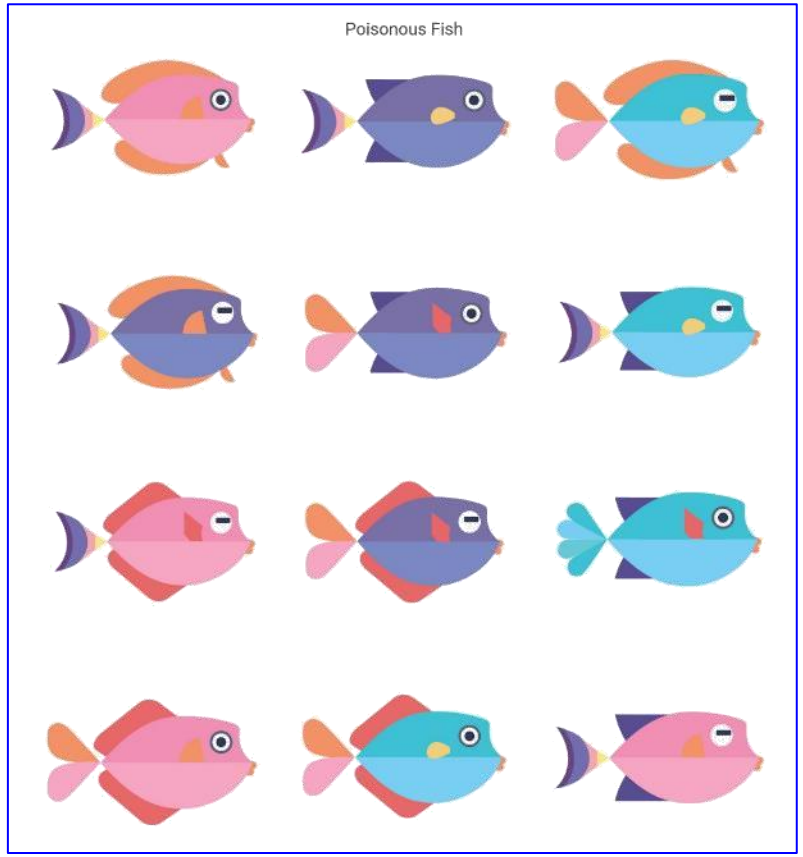
- Applying your fitted data mining models
  - Make comparison with students' prediction or measurement
  - Eg.: To predict the distance a paper airplane will fly

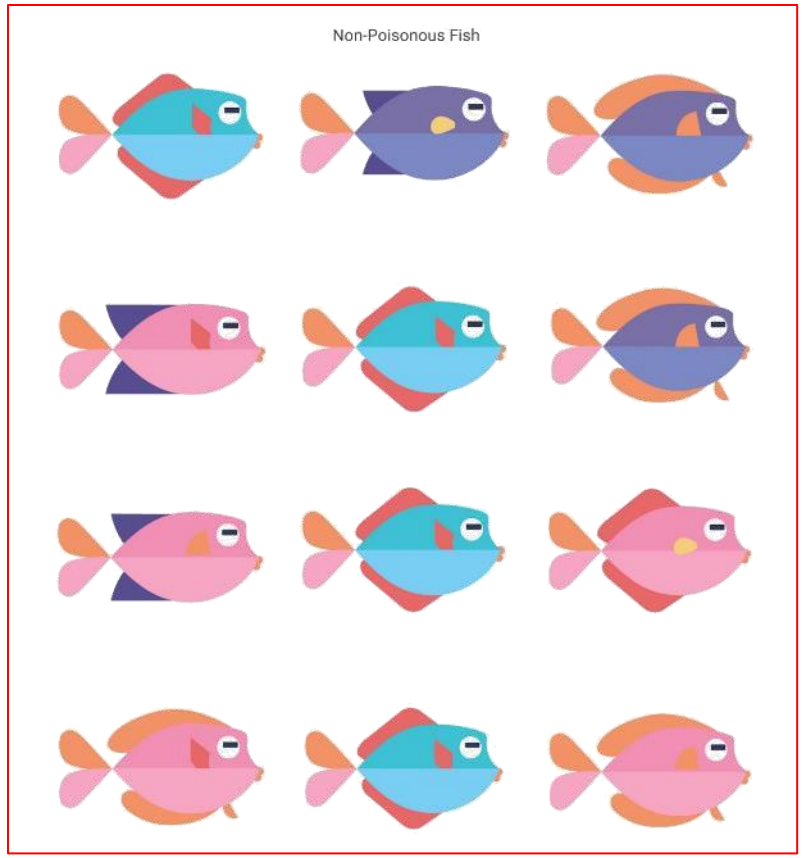
- Applying online data mining tools
  - Teach students to apply online tools to build models or use pre-trained models
  - Eg.: Teachable Machine, Code.org, ChatGPT

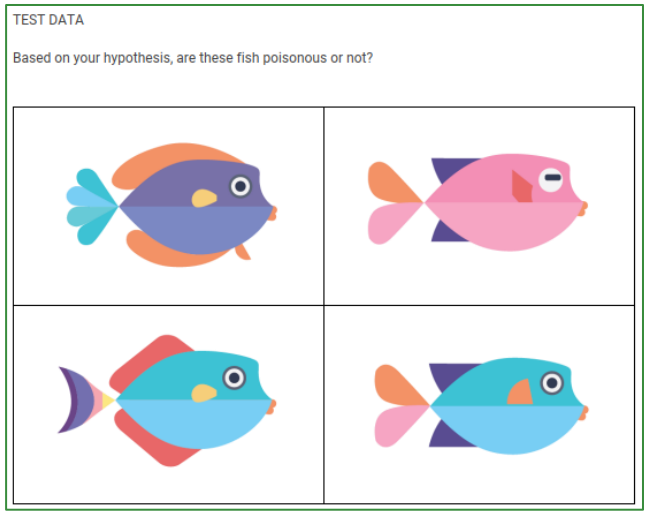
- Applying the results from your fitted data mining models
  - Eg.: Making a healthy burger, identifying the key factors for an improved fitness, focusing on food intake characteristics like weight, calories, salt content, oil content, and overall nutritional value, etc

STEM Lesson: Poisonous Fish Classification

- You are environmental scientists tasked with classifying fish as poisonous or non-poisonous. Can you come up with decision rules to determine if a fish is poisonous?
- Apply your rules to the test data and make prediction on whether these new fish are poisonous or non-poisonous.







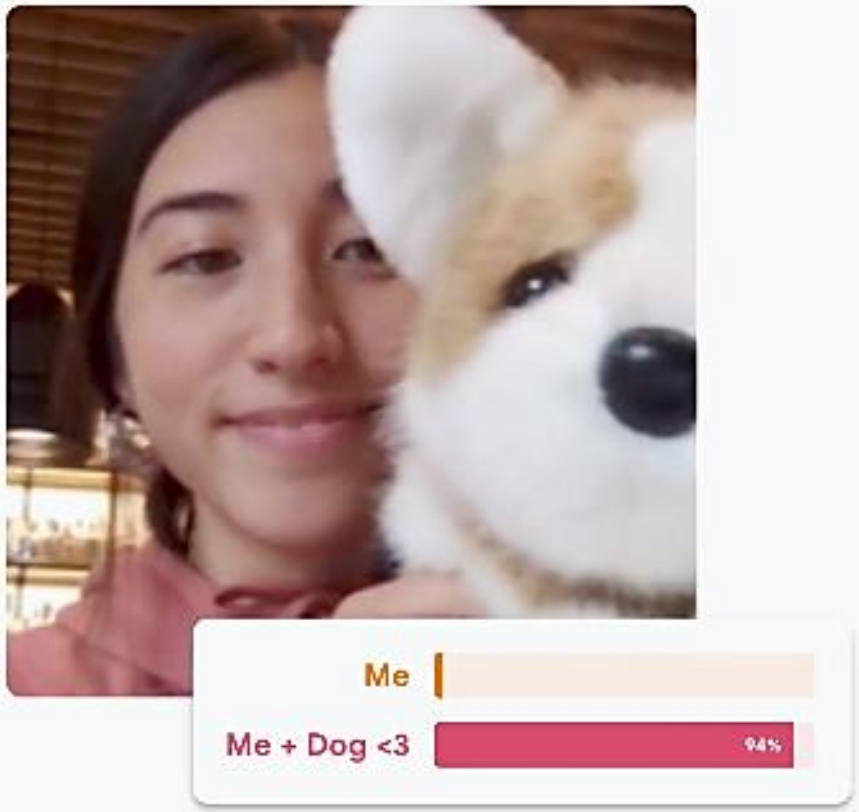
Teachable Machine

Teachable Machine

Train a computer to recognize your own images, sounds, & poses.

A fast, easy way to create machine learning models for your sites, apps, and more – no expertise or coding required.

Get Started



https://teachablemachine.withgoogle.com/

Example: Predicting the Quality of Mangoes

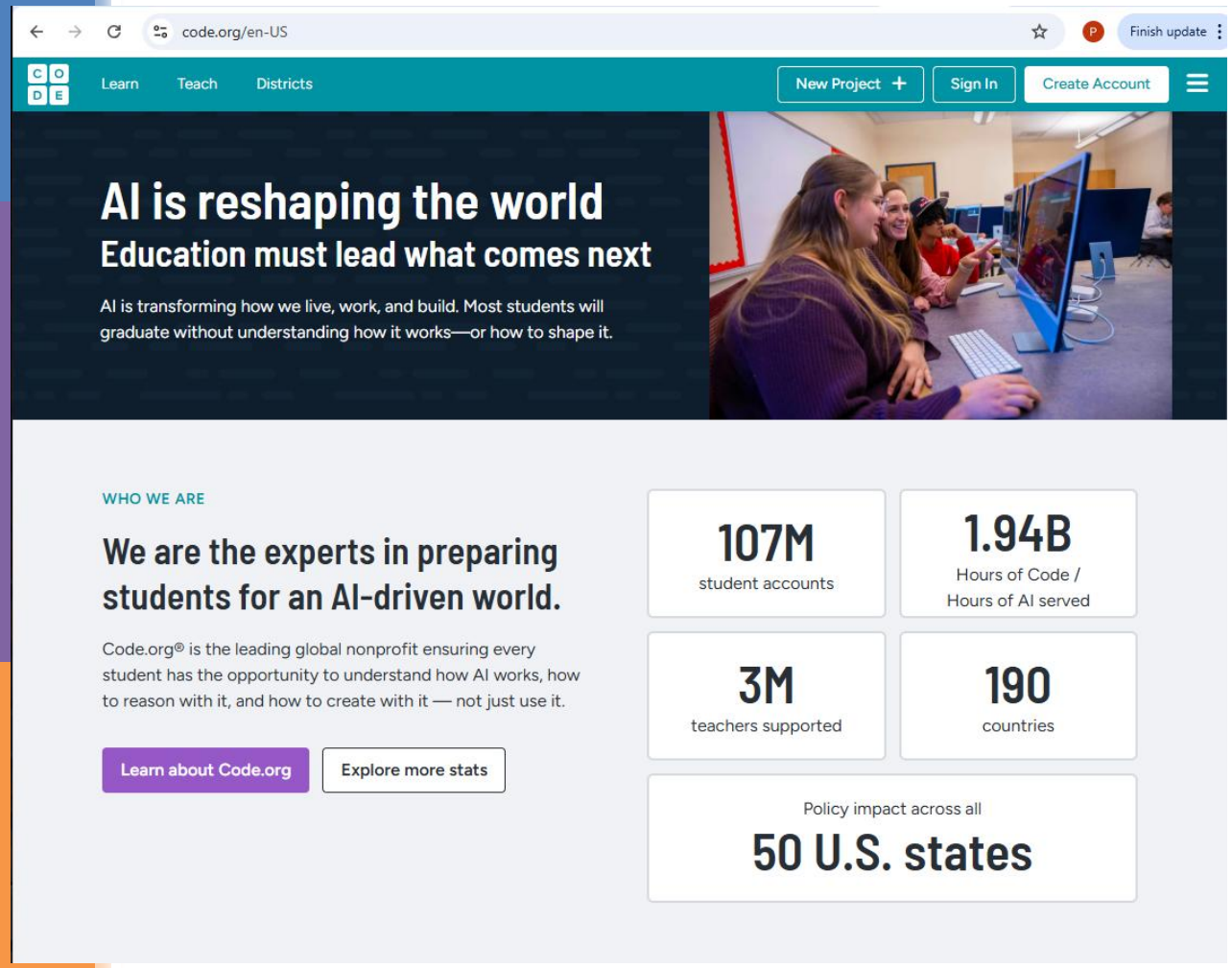
1. Define the quality of a mango (eg. Grades A (sweetest and most delicious), B and C)
2. Students collect samples of mangoes and record the colors of mango skin, texture of skin, softness, …, and take photos
3. The ground truth quality is based on the consensus of the teachers' sensory testing

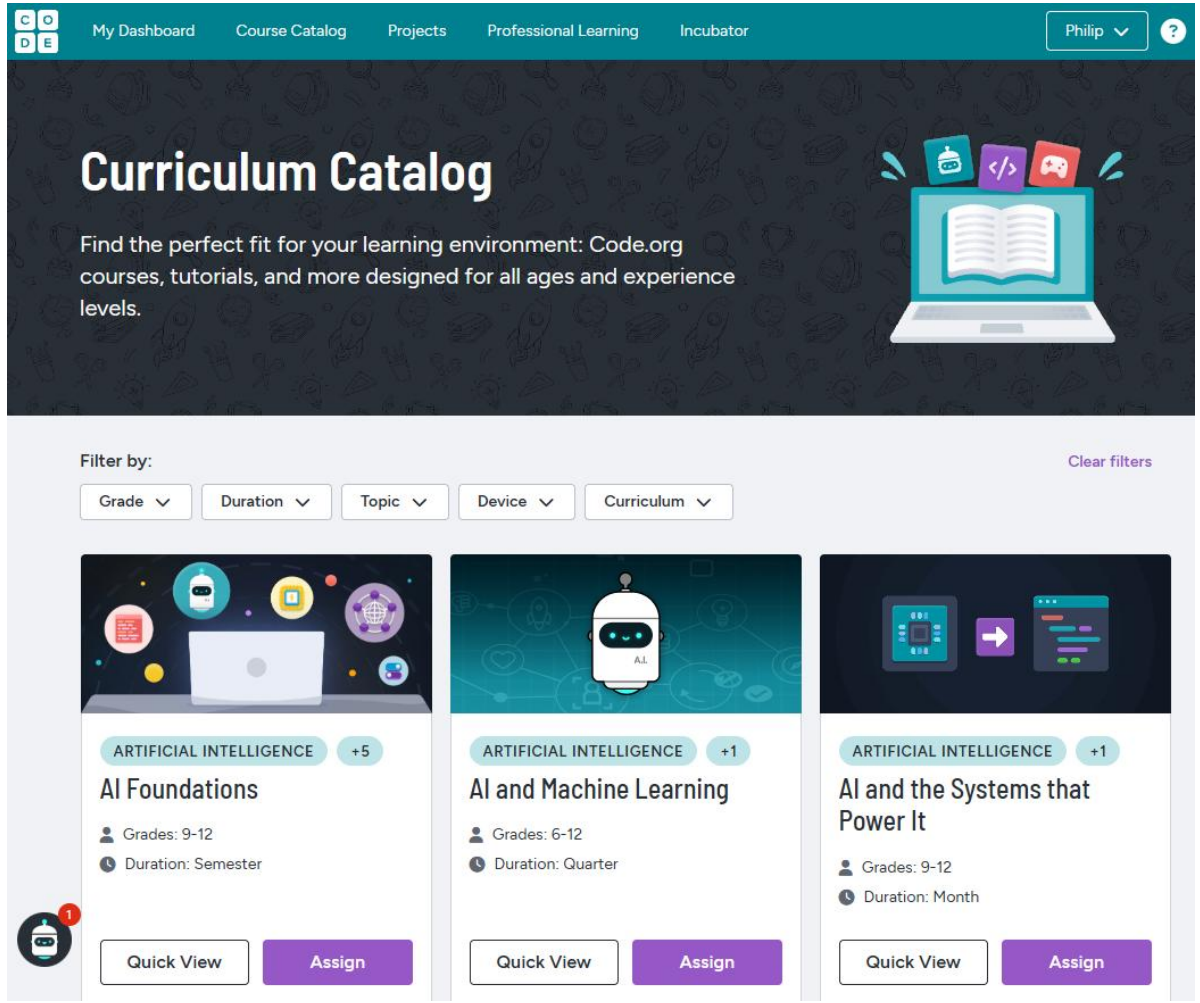
STEM Activities: Tasting game; Data visualization; ML or AI models

Sakulkueakulsuk, B., et al. (2018). Kids making AI: Integrating Machine Learning, Gamification, and Social Context in STEM Education. In *Proceedings of the 2018 IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE)*, 1005-1010.

Code.org

code.org





Code.org

code.org

AI for Oceans

Printing Options | + Assign to sections

View page as: Student | Teacher

Learn how AI and machine learning can be used to address world problems.

View calendar

Lesson 1: AI for Oceans

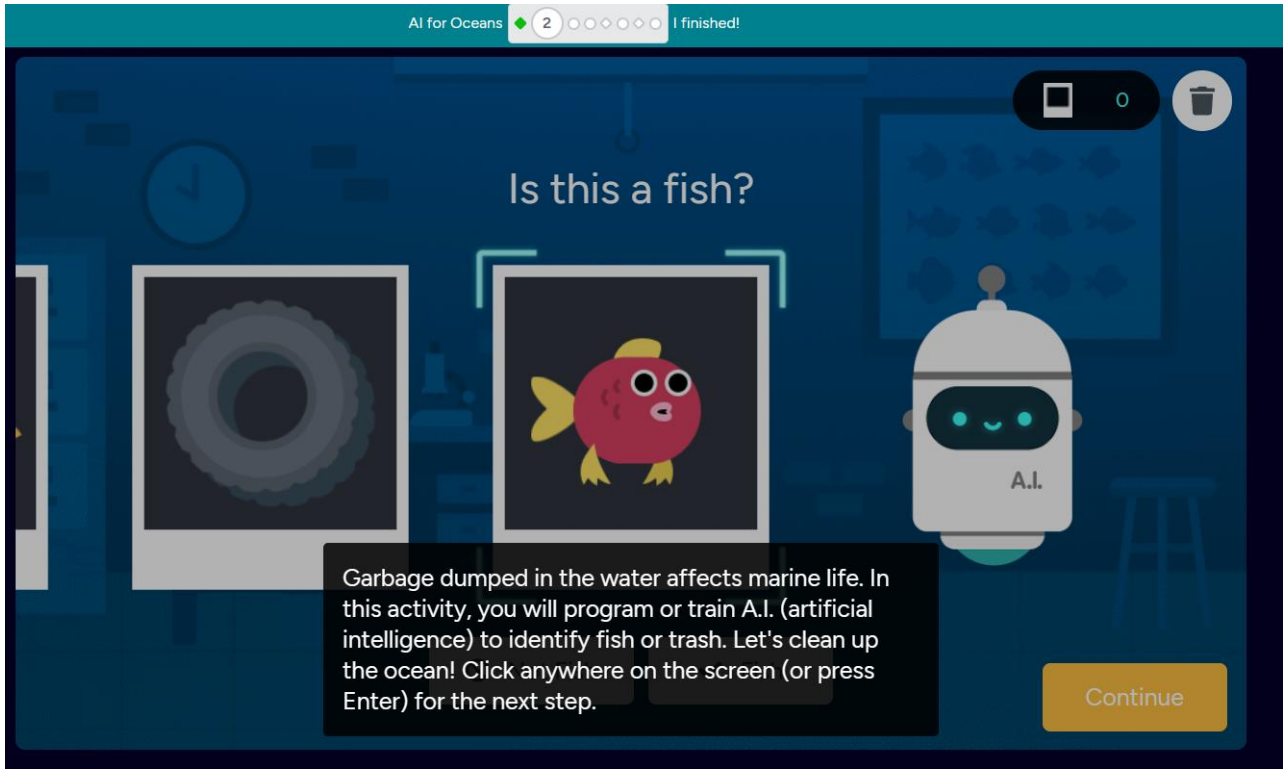
In this lesson, learners of all ages get an introductory experience with coding and computer science in a safe, supportive environment. First, students classify objects as either "fish" or "not fish" to attempt to remove trash from the ocean. Then, students will need to expand their training data set to include other sea creatures that belong in the water. In the second part of the activity, students will choose their own labels to apply to images of randomly generated fish. This training data is used for a machine learning model that should then be able to label new images on its own.

- 1 — AI Machine Learning - Video
- 2-4 — Train an AI to Clean the Ocean (2, 3, 4)
- 5 — AI: Training Data & Bias - Video
- 6 — Using Training Data
- 7 — AI: Impact on Society - Video
- 8 — Teach AI a New Word

View Lesson Plan

Send to students

Rate this Lesson



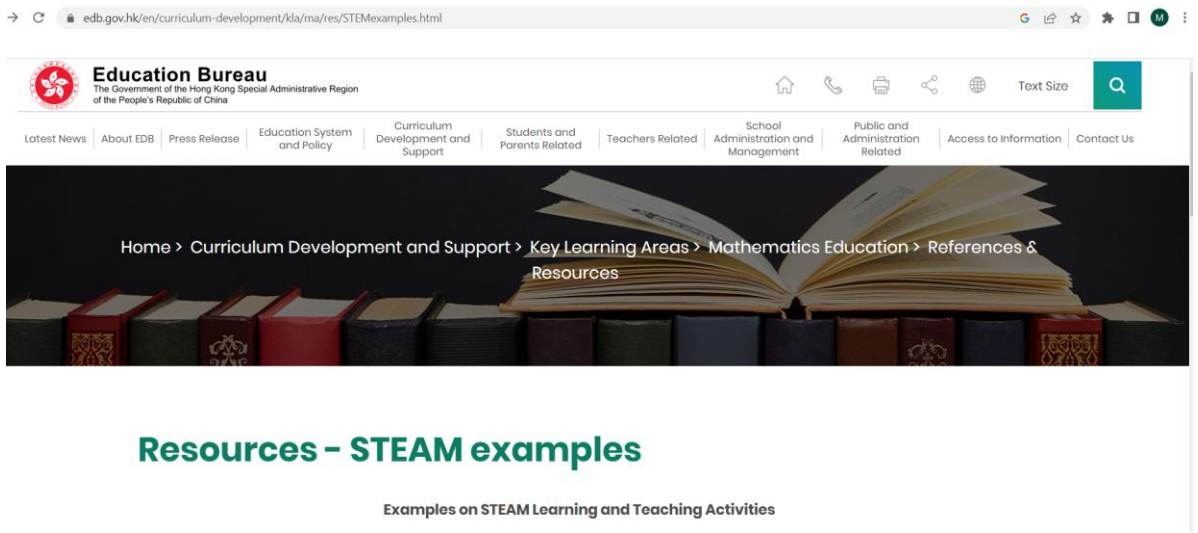
Class Activity 2:
Commenting on STEM Lesson Plans

1. State the objective(s) of the STEM lesson plan
2. State how data mining is used in this lesson plan
3. Comment on whether the time spent on each STEM activity is sufficient
4. Comment on the level of difficulty suitable for the targeted students

Commenting on STEM Lesson Plans

- STEM 1: Designing an Obesity-friendly Burger
- STEM 2: Unearthing Growth Secrets for a Thriving Greenhouse in STEM Education
- STEM 3: Revealing the Secret of Your Stress
- STEM 4: Zookeepers

Online STEM Resources



- EDB's STEAM Resources – https://www.edb.gov.hk/en/curriculum-development/kla/ma/res/STEMexamples.html https://www.edb.gov.hk/en/curriculum-development/kla/technology-edu/steam/index.html
- SPACE Foundation STEM Lesson Plans – https://discoverspace.org/stem-lesson-plans/
- daily STEM – http://dailystem.com/news/
- PBS Kids Design Squad – https://pbskids.org/designsquad/build/
- STEM Teaching Tools – http://stemteachingtools.org/
- Edutopia STEM Resources – https://www.edutopia.org/stem-to-steam-resources
- PBS – Engineering Design – https://wv.pbslearningmedia.org/subjects/engineering--technology/engineering-design-and-practices/
- Green Bank Observatory – https://greenbankobservatory.org/try-it-at-home
- PBL for Remote Learning – https://www.pblworks.org/pbl-remote-learning
- Teach Engineering – https://www.teachengineering.org/curriculum/browse?collection=Activities
- UN Sustainable Development Goals – https://www.un.org/sustainabledevelopment/sustainable-development-goals

THANK YOU, STEM TEACHERS!

https://twitter.com/seccardona/status/1600953044312936448